LECTURE – CONDITIONAL STATEMENTS
OOP (JAVA)

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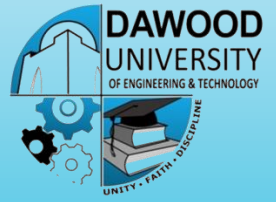
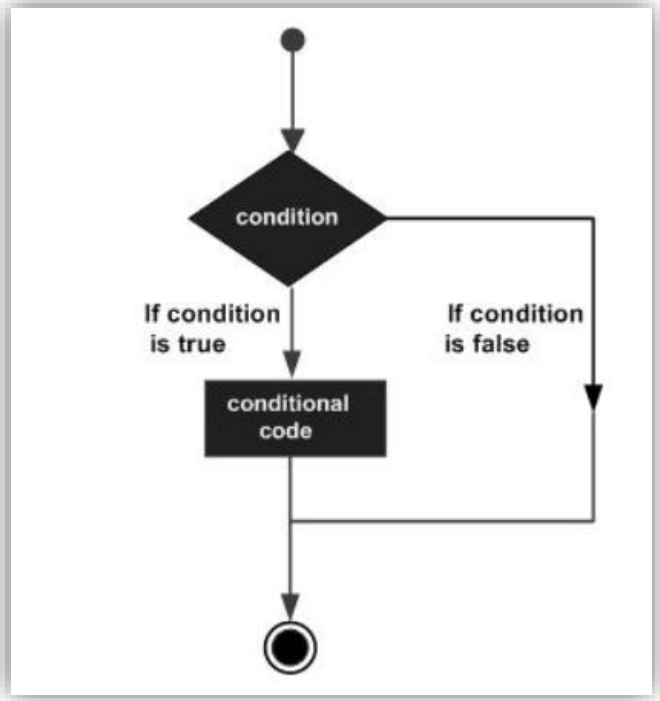
AGENDA



- Operators
- Types of Operator
- Keywords

Conditional Statements

- Java supports the usual logical conditions from mathematics:
  - **Less than: a < b**
  - **Less than or equal to: a <= b**
  - **Greater than: a > b**
  - **Greater than or equal to: a >= b**
  - **Equal to a == b**
  - **Not Equal to: a != b**



- can use these conditions to perform different actions for different decisions.

The if Statement

- Use the if statement to specify a block of Java code to be executed if a condition is true.
  - **Syntax**

```
if (condition) {
  // block of code to be executed if the condition is true
}
```

  - **Example**

```
if (20 > 18) {
  System.out.println("20 is greater than 18");
}
```



```
20 is greater than 18
```

The if Statement

- Use the if statement to specify a block of Java code to be executed if a condition is true.

```
int x = 20;
int y = 18;
if (x > y) {
  System.out.println("x is greater than y");
}
```

Example:

```
public class Main {
  public static void main(String[] args) {
    int x = 20;
    int y = 18;
    if (x > y) {
      System.out.println("x is greater than y");
    }
  }
```

Result:

```
x is greater than y
```

The else Statement

- Use the else statement to specify a block of Java code to be executed if a condition is false.
- Syntax

```
if (condition) {
  // block of code to be executed if the condition is true
} else {
  // block of code to be executed if the condition is false
}
```

The else Statement

- Use the else statement to specify a block of Java code to be executed if a condition is false.

```
int time = 20;
if (time < 18) {
  System.out.println("Good day.");
} else {
  System.out.println("Good evening.");
}
```

Example:

```
public class Main {
  public static void main(String[] args) {
    int time = 20;
    if (time < 18) {
      System.out.println("Good day.");
    } else {
      System.out.println("Good evening.");
    }
  }
```

Result:

```
Good evening.
```

The else if Statement

- Use the else if statement to specify a new condition if the first condition is false.
- Syntax

```
if (condition1) {
  // block of code to be executed if condition1 is true
} else if (condition2) {
  // block of code to be executed if the condition1 is false and condition2 is true
} else {
  // block of code to be executed if the condition1 is false and condition2 is false
}
```

The else if Statement

- Use the else if statement to specify a new condition if the first condition is false.

```
int time = 22;
if (time < 10) {
  System.out.println("Good morning.");
} else if (time < 20) {
  System.out.println("Good day.");
} else {
  System.out.println("Good evening.");
}
```

Example:

```
public class Main {
  public static void main(String[] args) {
    int time = 22;
    if (time < 10) {
      System.out.println("Good morning.");
    } else if (time < 20) {
      System.out.println("Good day.");
    }  else {
```

Result:

```
Good evening.
```

The short hand if else Statement (Ternary Operator)

- It consists of three operands.
- It can be used to replace multiple lines of code with a single line.
- It is often used to replace simple if else statements

Example

```
variable = (condition) ? expressionTrue :  expressionFalse;
```

The short hand if else Statement (Ternary Operator)

- It can be used to replace multiple lines of code with a single line.

```
int time = 20;
if (time < 18) {
  System.out.println("Good day.");
} else {
  System.out.println("Good evening.");
}
```

```
Good evening.
```

Example:

```
int time = 20;
String result = (time < 18) ? "Good day." : "Good evening.";
System.out.println(result);
```

```
public class Main {
  public static void main(String[] args) {
    int time = 20;
    String result;
    result = (time < 18) ? "Good day." : "Good evening.";
    System.out.println(result);
```

Result:

```
Good evening.
```

The Switch Statement

- Use the switch statement to select one of many code blocks to be executed.
- Syntax

```
switch(expression) {
  case x:
    // code block
    break;
  case y:
    // code block
    break;
  default:
    // code block
}
```

The switch expression is evaluated once.

The value of the expression is compared with the values of each case.

If there is a match, the associated block of code is executed.

- The break and default keywords are optional, and will be described later.

The Switch Statement

Example:

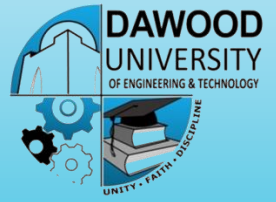
```
int day = 4;
switch (day) {
  case 1:
    System.out.println("Monday");
    break;
  case 2:
    System.out.println("Tuesday");
    break;
  case 3:
    System.out.println("Wednesday");
    break;
  case 4:
    System.out.println("Thursday");
    break;
  case 5:
    System.out.println("Friday");
    break;
  case 6:
    System.out.println("Saturday");
    break;
  case 7:
```

```
public class Main {
  public static void main(String[] args) {
    int day = 4;
    switch (day) {
      case 1:
        System.out.println("Monday");
        break;
      case 2:
        System.out.println("Tuesday");
        break;
      case 3:
        System.out.println("Wednesday");
        break;
      case 4:
        System.out.println("Thursday");
        break;
      case 5:
        System.out.println("Friday");
        break;
      case 6:
        System.out.println("Saturday");
        break;
      case 7:
        System.out.println("Sunday");
        break;
```

Result:

```
Thursday
```

The Switch Statement



The break Keyword

- When it reaches a break keyword, it breaks out of the switch block.
- This will stop the execution of more code and case testing inside the block.
- When a match is found, and the job is done, it's time for a break. There is no need for more testing.
- Syntax

```
switch(expression) {
  case x:
    // code block
    break;
  case y:
    // code block
    break;
  default:
    // code block
}
```

The Switch Statement

The default Keyword

- The default keyword specifies some code to run if there is no case match:.

Example:

```
int day = 4;
switch (day) {
  case 6:
    System.out.println("Today is Saturday");
    break;
  case 7:
    System.out.println("Today is Sunday");
    break;
  default:
    System.out.println("Looking forward to the Weekend");
}
```



```
public class Main {
  public static void main(String[] args) {
    int day = 4;
    switch (day) {
      case 6:
        System.out.println("Today is Saturday");
        break;
      case 7:
        System.out.println("Today is Sunday");
        break;
      default:
        System.out.println("Looking forward to the Weekend");
    }
  }
}
```

Result:

```
Looking forward to the Weekend
```

Thanks ☺